CSCI 8820 Computer Vision and Pattern Recognition

Assignment 2, Due Friday, February 21, 2025 by 11.59pm (23:59 EST)

For the binary image **B** generated in the previous assignment:

1. Implement an algorithm to determine the Medial Axis (skeleton) for each object (component) in the image. Display the resulting Medial Axis image **M** as a binary image wherein the axis (skeleton) pixels are 1 and the rest are 0. Note that **M** by itself is not a binary image (i.e., it retains the distance transform value at each skeleton pixel).

2. Implement an algorithm to reconstruct the original binary image from the Medial Axis image **M**. Provide an outline of the reconstruction algorithm in high-level pseudo-code. Display the reconstructed binary image $\mathbf{B_R}$**. Note:** the reconstruction should be lossless.

<u>Submit the assignment as a **single PDF file** to the specified ELC dropbox</u>. The PDF file should include the following:

1. An outline of the reconstruction algorithm in high-level pseudo-code.
2. A *well documented* hardcopy of the source code for the computation of the Medial Axis and the reconstruction of the original binary image from the Medial Axis.
3. Hardcopies of the images **B**, **M** and $\mathbf{B_R}$**.**
4. Comments on the results obtained.

NOTE: Please make sure that the PDF file contains your name and UGA ID Number on the first page. Only submissions received in the ELC dropbox by the stated deadline will be considered. Submissions received after the deadline via email (or any other means) will **not** be graded and will be assigned a zero grade. <u>The assignment should be submitted as a single PDF file only.</u>